A Further Note on the Statistical Meaning of Photon Number in a Thermal-Equilibrium Radiation System

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Abstract

This paper is a further supplement to the third part of the previous paper, concerning the statistical meaning of photon number in a thermal-equilibrium radiation system. It further points out that, from the perspective of structure ontology, the identity of the photon in statistical mechanics is that of a count.

Preliminary Remarks

In the third part of the previous paper, on the statistical meaning of photon number in a thermal-equilibrium radiation system [1], it was pointed out that the occupation number and the photon number are equivalent in statistical meaning, and that both belong to statistical quantities. On this basis, the present paper further explores the position of the photon in statistical physics from the perspective of structure ontology.

It is generally considered that the light has wave-particle duality, and that it is grasped through physical pictures such as particle or wave. But within the framework of structure ontology in statistical mechanics, in what identity does the photon appear? This paper points out that, in structure ontology, the photon appears as a count in statistics.

Therefore, within the structure ontology of statistical mechanics, the identity of the photon is no longer that of a particle, but that of a count in statistics. From this, we can naturally see that structure ontology, the central concern is the macroscopic statistical structure, rather than the microscopic details of individuals.

Discussion

In statistical mechanics, if structure ontology is taken as the point of departure, then under this description, what exactly is the identity of the photon?

On the one hand, according to the previous paper, the photon number is a statistical quantity. On the other hand, in the expression of the black-body radiation law, it is not natural to imply the second Lagrange multiplier introduced by the constraint condition on the total photon number. Combining these two aspects, we may regard the photon as existing in statistics as a count.

Extending from the viewpoint, several points can be obtained:

First, here, the light is no longer grasped a particle or as a wave in the usual physical sense, and the photon is interpreted as a count in statistics. Therefore, the photon no longer has the physical “existence” of a particle. Thus, there is no need to describe photons through a concrete physical picture; for example, there is no need to ask whether, in a thermal equilibrium radiation system, the total number of photons is conserved. This is not a question that needs to be judged, because the photon itself is, in the structure description, a statistical manifestation of the distribution of the system's energy.

Second, in statistical mechanics that takes structure as its point of departure, the emphasis is placed on the importance of the overall structure of energy occupation rather than on the dynamical behavior of individuals. When the total energy within a system changes, it is first manifested as an adjustment of the structure of energy occupation, and on the macroscopic level, this change is represented through a change in temperature.

Third, structure ontology is not concerned with the detailed issues of individuals, such as interactions between photons or the current state of a single photon. It is concerned only with how energy is distributed over phase cells so that the entropy of the system reaches its maximum. Any macroscopic quantity is the result of statistics and does not involve the microscopic causal relations of individuals.

References

[1] Bo Yang. A discussion of phase cells as the basic descriptive units of structure in statistical mechanics. 2026. DOI:10.5281/zenodo.20237784.